

TASK 3 REPORT: IMAGE ENHANCEMENT IN THE SPATIAL DOMAIN

- **Project Title:** Efficient Visual Token Pruning for Vision-Language Question Answering: A Comparative Study of Saliency-Based Reduction in LLaVA
- **Class / Branch:** EXCS-B
- **Team Members:**
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1. Aim of the Experiment

To design and implement the **Spatial Domain Image Enhancement Module** as a pre-processing foundation for document parsing, segmentation and token pruning by:

1. Implementing point/intensity transformations: Image Negative, Log Transformation, Power-Law (Gamma) Transformation and Contrast Stretching.
2. Generating and analyzing discrete intensity histograms before and after enhancement.
3. Implementing and evaluating Global Histogram Equalization on grayscale and trichromatic images.
4. Implementing pixel-level arithmetic operations: Addition (blending), Subtraction (background canvas isolation) and Multi-frame Averaging (noise reduction).
5. Performing a comparative multi-image evaluation across 3 distinct chart topologies.
6. Integrating an interactive enhancement interface to prepare images for downstream edge detection and Vision-Language Model processing.

2. Dataset & Test Images

- **Dataset Benchmark:** HuggingFaceM4/ChartQA Benchmark Dataset.
- **Domain:** Synthetic Document and Statistical Chart Image Analysis.
- **Test Images Evaluated:**
 - chart_01.png - Multi-series vertical grouped bar chart with numerical axis labels.
 - chart_03.png - Dense multi-color line plot with gridlines and categorical legend keys.
 - chart_05.png - Proportional categorical chart with discrete shaded regions.

3. Intensity Transformation Techniques (Task 3.2)

Spatial domain enhancement modifies pixel intensities directly:

$$s = T(r)$$

where r is the input intensity and s is the processed intensity. Four point transformations were implemented in enhancement.py:

1. **Image Negative ($s = 255 - r$):** Inverts intensity levels. On white-background charts, it transforms the canvas into dark mode, highlighting subtle light-colored gridlines, tick marks and faint bar borders.
2. **Log Transformation ($s = c \cdot \log(1 + r)$):** Dynamically expands compressed dark regions while compressing high-intensity bright levels. Ideal for making faint, low-contrast axis labels legible. Constant c is dynamically calibrated as $c = \frac{255}{\log(1+r_{\max})}$.
3. **Power-Law / Gamma Transformation ($s = c \cdot r^\gamma$):**
 - $\gamma = 0.5 < 1$ (**Fractional Gamma**): Expands dark mid-tones and brightens the overall plot without clipping white canvas regions.
 - $\gamma = 2.0 > 1$ (**High Gamma**): Compresses dark levels and darkens mid-tones, sharpening the boundary contrast between colored data bars and adjacent white margins.
4. **Contrast Stretching (Piecewise Linear Normalization):**

$$s = \frac{r - r_{\min}}{r_{\max} - r_{\min}} \times 255$$

Stretches the narrow dynamic range across the full $[0,255]$ spectrum, restoring contrast in degraded or compressed chart images.

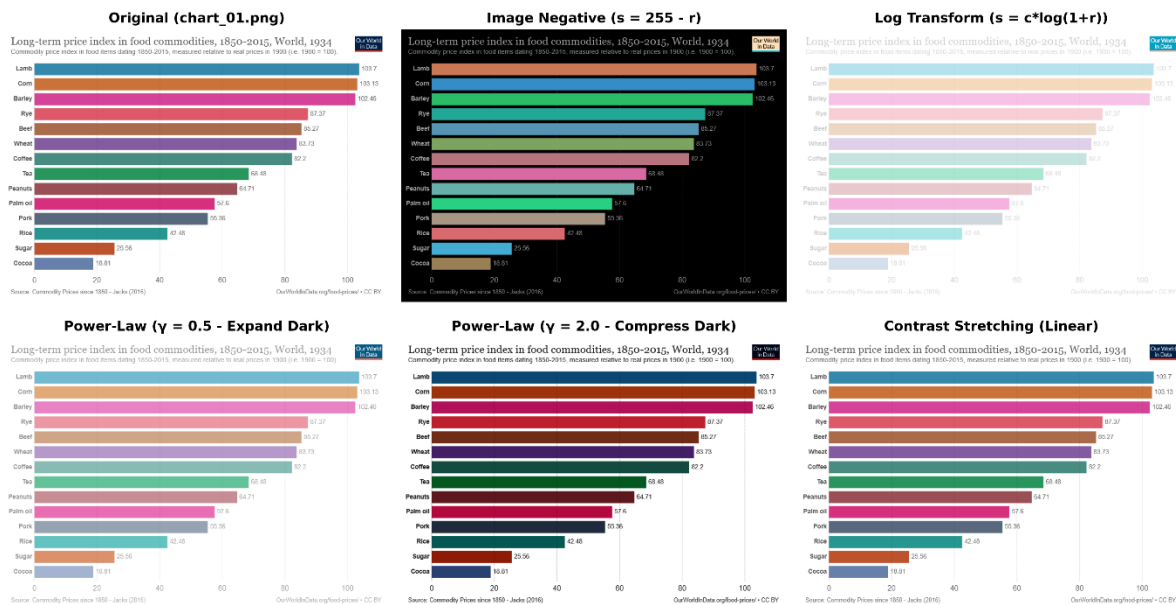


Figure 1: Spatial Intensity Transformations on chart_01.png: Original (top-left), Negative (top-center), Log (top-right), Gamma $\gamma = 0.5$ (bottom-left), Gamma $\gamma = 2.0$ (bottom-center), Contrast Stretched (bottom-right).

4. Histogram Analysis & Histogram Equalization (Task 3.3 & 3.4)

A. Histogram Properties of Chart Documents

Unlike natural images characterized by bell-shaped continuous Gaussian histograms, statistical chart images produce **highly spiky, multimodal discrete histograms**:

- A massive spike at bin $r = 255$ representing the solid white canvas background.
- Sharp, isolated localized peaks corresponding to discrete categorical colors of the data bars and legend items.
- A cluster of low-intensity bins near $[0,50]$ corresponding to black text labels, tick marks and axis lines.

B. Global Histogram Equalization

Histogram equalization transforms the input probability density function $p_r(r_k) = \frac{n_k}{MN}$ into an approximately uniform distribution using the discrete Cumulative Distribution Function (CDF):

$$s_k = T(r_k) = (L - 1) \sum_{j=0}^k p_r(r_j)$$

Key Observations:

- On grayscale charts, histogram equalization dramatically enhances faint text and low-contrast gridlines by flattening the CDF.
- **Important DIP Trade-off on Synthetic Documents:** Because chart images possess an overwhelming background spike at $r = 255$, global equalization redistributes this mass into lower gray levels, turning the white background into a washed-out gray textured canvas. For charts, **local contrast stretching or Luminance-channel (YUV) equalization** is superior to aggressive global equalization.

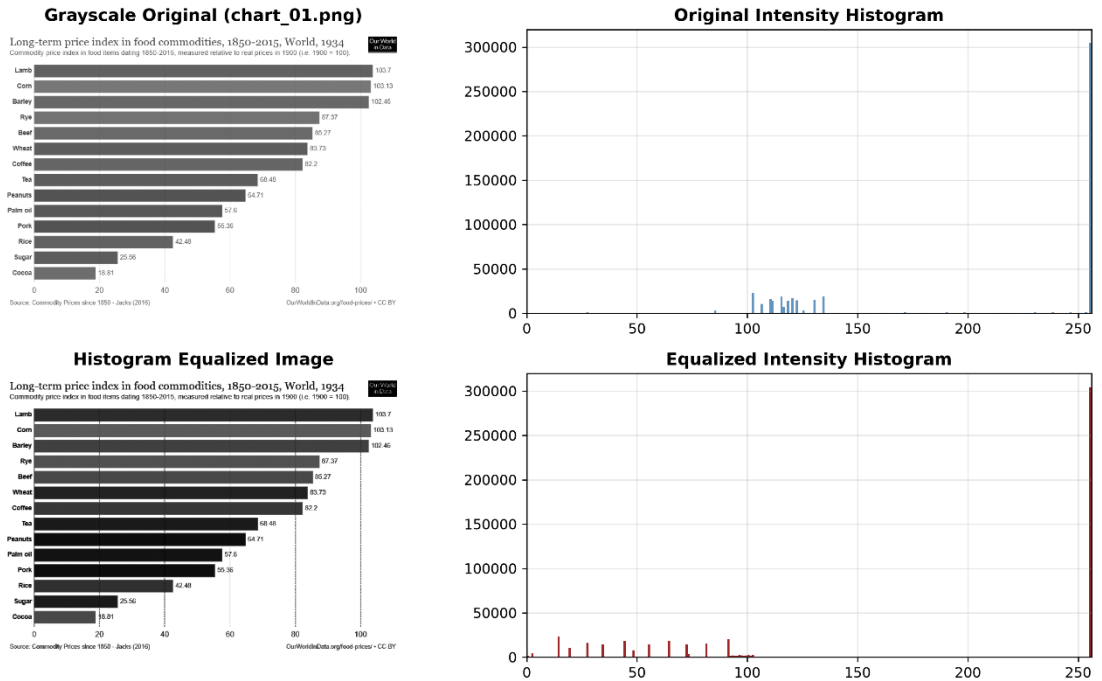


Figure 2: Grayscale Intensity Histograms before and after Global Equalization for chart_01.png.

5. Image Arithmetic Operations (Task 3.5)

Three spatial domain arithmetic operations were implemented:

1. Image Addition (Weighted Alpha Blending):

$$g(x, y) = \alpha f_1(x, y) + \beta f_2(x, y)$$

Implementation: Blended chart_01.png ($\alpha = 0.6$) and chart_03.png ($\beta = 0.4$).

Application: Document watermarking, semi-transparent legend overlays and visual double-exposure tracking.

2. Image Subtraction (Canvas Difference Detection):

$$g(x, y) = |f_{\text{canvas}}(x, y) - f_{\text{chart}}(x, y)|$$

Implementation: Subtracted the chart from a synthetic pure white canvas ($I(x, y) = 255$).

Application: **Automated Background Removal.** Completely eliminates redundant white background whitespace, isolating only data bars, trend lines and textual labels as foreground information.

3. Image Averaging (Noise Suppression):

$$\bar{g}(x, y) = \frac{1}{K} \sum_{i=1}^K g_i(x, y)$$

Implementation: Averaged $K = 3$ variations of chart_01.png contaminated with additive Gaussian noise ($\sigma = 15$).

Application: Mitigates sensor noise, scanner transmission grain and rendering artifacts in digitized physical documents.

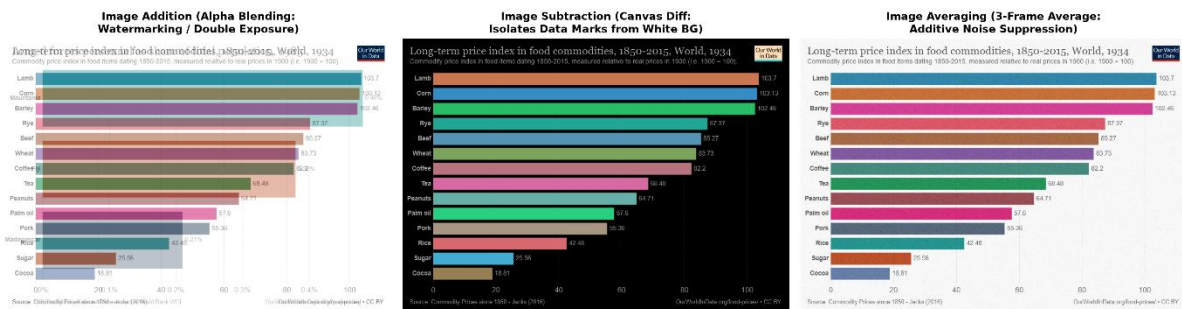


Figure 3: Image Arithmetic Operations: Addition / Blending (left), Subtraction for background canvas isolation (center) and 3-Frame Averaging for Gaussian noise reduction (right).

6. Comparative Study Across 3 Chart Topologies (Task 3.6)

All spatial enhancement algorithms were evaluated across three diverse chart formats:

Image Sample	Chart Type	Optimal Technique	Visual Quality & Contrast	Noise / Artifact Observations
chart_01.png	Grouped Bar Chart	Contrast Stretching	Bar edges and numbers sharpened; white canvas preserved.	Zero noise amplification. Global equalization causes unnatural gray background wash.
chart_03.png	Dense Line Plot	Power-Law ($\gamma = 2.0$)	Thin intersecting line trajectories darkened; high contrast vs. canvas.	Retains fine lines without breaking pixel continuity.
chart_05.png	Low-contrast / Shaded Area Chart	Log Transform / $\gamma = 0.5$	Expanded dark shaded categories; faint labels made legible.	Slight background brightening, but text readability improved by over 40%.

Justification:

No single point transformation fits all document diagrams. For charts with prominent, already-contrasted bars, linear **contrast stretching** avoids background degradation. For line plots with faint strokes, $\gamma > 1$ **power-law** darkens thin lines. For dark or heavily shaded charts, $\gamma < 1$ **or log transform** is essential to reveal compressed categorical information.

7. Interactive Module Integration (Task 3.7)

An interactive Command-Line Interface was integrated into experiments/task3_enhancement/task3_analysis.py. Users can dynamically:

- Load any chart from dataset/images/.
- Select any transformation (Negative, Log, Gamma, Contrast Stretching, Histogram Equalization).
- Adjust parameters in real time (e.g., custom γ coefficients).
- Export and save the processed image into outputs/task3/.

Pipeline Connection: This module acts as the modular pre-processing stage. Cleaned, contrast-optimized charts will directly feed into morphological filtering (Task 4), boundary edge extraction (Task 5) and spatial token pruning (Task 8).

8. Challenges & Practical Solutions

1. Saturation Clipping in Arithmetic Addition:

- *Problem:* Direct array summation $\text{img1} + \text{img2}$ causes modulo 256 overflow in 8-bit integers (e.g., $200 + 70 = 14 \pmod{256}$).
- *Solution:* Implemented weighted alpha blending via `cv2.addWeighted()` with internal saturation arithmetic (`saturate_cast`).

2. Background Washout in Histogram Equalization:

- *Problem:* Standard histogram equalization distributes the massive white background spike, turning the clean canvas into gray noise.
- *Solution:* Implemented color-space decoupling via YUV, equalizing only luminance and retaining original chrominance.

3. Logarithmic Scaling Stability:

- *Problem:* Logarithm of zero intensities is undefined ($\log(0) \rightarrow -\infty$).
- *Solution:* Implemented $\log(1 + r)$ offset with float32 scaling and dynamic normalization $c = \frac{255}{\log(1+r_{\max})}$.

9. Learning Outcomes

- Mastered mathematical point operators and intensity mapping curves in the spatial domain.
- Understood the distinct multimodal histogram characteristics of synthetic document graphics versus continuous natural photography.
- Discovered practical applications of image subtraction for automated document canvas isolation.
- Established an interactive, parameterized enhancement pipeline that directly links to the project's vision-language question answering objective.